

The diagram illustrates the sulfur cycle with the following components and reactions:

- Sulfur Species (in red):** HS^- , $\text{S}(0)$, SO_3^{2-} , $\text{S}_2\text{O}_3^{2-}$, $\text{S}_4\text{O}_6^{2-}$, SO_4^{2-} , and **APS**.
- Enzymes and Pathways:**
 - Top Pathway:** $\text{HS}^- \xrightarrow{\text{FccAB/Sqr}} \text{S}(0)$ (solid arrow); $\text{S}(0) \xrightarrow{\text{SOR}} \text{HS}^-$ (dashed arrow).
 - Left Pathway:** $\text{SO}_3^{2-} \xrightarrow{\text{PhsABC}} \text{S}_2\text{O}_3^{2-}$ (solid arrow); $\text{S}_2\text{O}_3^{2-} \xrightarrow{\text{PhsABC}} \text{SO}_3^{2-}$ (solid arrow).
 - Central Pathway:** $\text{S}_2\text{O}_3^{2-} \xrightarrow{\text{TsdA/DoxDA}} \text{S}_4\text{O}_6^{2-}$ (solid arrow); $\text{S}_4\text{O}_6^{2-} \xrightarrow{\text{Ttr/Otr}} \text{S}_2\text{O}_3^{2-}$ (dashed arrow).
 - Right Pathway:** $\text{S}(0) \xrightarrow{\text{SoxB}} \text{SO}_3^{2-}$ (solid arrow); $\text{SO}_3^{2-} \xrightarrow{\text{SOR}} \text{S}(0)$ (dashed arrow).
 - Bottom Pathway:** $\text{SO}_3^{2-} \xrightarrow{\text{SoxYZXAB}} \text{SO}_4^{2-}$ (dashed arrow); $\text{S}_2\text{O}_3^{2-} \xrightarrow{\text{SoxYZXABCD}} \text{SO}_4^{2-}$ (dashed arrow); $\text{S}_4\text{O}_6^{2-} \xrightarrow{\text{SoxB}} \text{SO}_4^{2-}$ (solid arrow); $\text{SO}_4^{2-} \xrightarrow{\text{SoxYZXAB}} \text{S}_2\text{O}_3^{2-}$ (dashed arrow).
 - APS Pathway:** $\text{APS} \xrightarrow{\text{Sat}} \text{SO}_4^{2-}$ (solid arrow); $\text{SO}_4^{2-} \xrightarrow{\text{AprAB}} \text{APS}$ (solid arrow).
 - Electron Flow:** DsrMKJOP is involved in the reduction of HS^- to $\text{S}(0)$ and the reduction of SO_3^{2-} to $\text{S}(0)$ (indicated by dotted arrows and e^-).
 - Other Enzymes:** PsrABC/ShyABCD , SudAB/Npsr/Nsr , SoxYZXAB , TetH , SoeABC , SorAB , SorT , sHdrAB1B2B3C1C2 , **DsrABCE**, **FHL**, and **Sdo** are also indicated in the diagram.

Dissimilatory sulfate reduction pathway

The diagram illustrates the dissimilatory sulfate reduction pathway, showing the flow of electrons and the conversion of various compounds across the cell membrane.

Periplasm:

- SO_4^{2-} is reduced to APS by the enzyme **Sat**.
- APS is reduced to SO_3^{2-} by the enzyme **AprBA**.

Membrane:

- QmoABC** is a transmembrane protein that facilitates the reduction of MQH_2 to **MQ**.
- DsrMKJOP** is a transmembrane protein that facilitates the reduction of MQH_2 to **MQ**.

Cytoplasm:

- SO_3^{2-} is reduced to HS^- by the enzyme **DsrBA**.
- HS^- is oxidized to $\text{S}(0)$ by the enzyme **DsrC**.
- $\text{S}(0)$ is oxidized back to SO_4^{2-} by the enzyme **DsrC**.

Electron Flow:

- Electrons (e^-) flow from **QmoABC** to **AprBA**.
- Electrons flow from **DsrMKJOP** to **DsrC**.

[Weissgerber et al. \(2014\)](#)

[Gwak et al. \(2022\)](#)

Periplasm

Membrane

Cytoplasm

Dissimilatory sulfide oxidation pathway